

PAPER

GraphyloVar: Predicting the impact of non-coding mutations using a multi-species sequence model

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Abstract

Motivation: Figuring out what genetic mutations do is a key problem for precision medicine. Tools like CADD, PhyloP, and PhastCons are useful, but they often look at each position in the genome in isolation. This means they can miss important information from the evolutionary history that connects different species. In this paper, we extend our previous model, Graphylo, to predict the effects of mutations. Our new model, named GraphyloVar, is different because it is built to directly utilize the phylogenetic tree that relates the species.

Methods: GraphyloVar is a deep learning model that considers both DNA sequence and evolutionary patterns from many species. It uses two main components: Graph Convolutional Networks (GCNs) to process the phylogenetic tree, and Transformer Encoders to extract features from the DNA sequences. With self-attention, the model learns to focus on important positions in the sequence and important relationships between species. We pre-train the model to predict allele frequencies in a population, which helps it learn about both the evolutionary history between species and the genetic variation within a species. The model has two outputs: one predicts the probability of each allele (A, C, G, T), and the other predicts the probability that the position is polymorphic.

Conclusion: By integrating deep learning with explicit phylogenetic input, GraphyloVar offers a powerful and complementary approach to mutation effect prediction. Unlike methods that rely solely on sequence conservation or single-species data, our method utilizes the full evolutionary history from many species to gain a better understanding of how a mutation might disrupt an important part of the genome. This helps us to better identify and prioritize important non-coding variants for future study.

Availability: Code and datasets are available at <https://github.com/DongjoonLim/GraphyloVar>

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Introduction

Understanding the functional consequences of genetic mutations is an important foundation for the advancement of precision medicine and the decoding of complex relationships between genotype and phenotype. Subtle alterations in the genome can have significant effects on phenotypic outcomes. Large-scale efforts, such as the 1000 Genomes Project and the ENCODE Project [??] have shown that the human genome is highly complex, with various regulatory components and many genetic variants whose functions are still not well understood. Although genome-wide association studies (GWAS) have identified numerous genetic variants associated with disease phenotypes, they often do not explain the precise molecular mechanisms through which specific mutations cause their effects [?]. Approximately 90% of disease-associated genetic variants are located in non-coding regions, with a likely impact on transcriptional or post-transcriptional regulation. However, the mechanisms through which these non-coding

variants influence gene expression, regulatory networks, and downstream cellular processes remain poorly understood.

To address this problem, researchers have turned to computational methods. One common strategy is to use measures of evolutionary conservation. Tools such as PhyloP and PhastCons were developed to quantify how much a specific position in the genome has been conserved across different species [??]. The idea is that mutations in highly conserved positions are more likely to be harmful. Following this logic, integrative tools, such as CADD were created to predict the functional impact of variants by combining these conservation scores with many other types of data, such as functional annotations [?].

More recently, a different class of deep learning models, such as DeepSEA and Enformer, have been developed [??]. They use long stretches of DNA sequence (typically human) to predict features like chromatin accessibility, histone marks, and gene expression. These models have shown to be useful

at capturing regulatory signals from the sequence. However, both approaches have limitations. The conservation-based methods use evolutionary context but often ignore local sequence patterns. On the other hand, the single-species deep learning models are powerful at using local sequence context but do not explicitly incorporate the broader evolutionary information that comes from comparing multiple species. This is a particular problem for understanding non-coding regions, where a combination of both types of information may be needed to find subtle evolutionary constraints.

Building on the above contrast between single-species sequence models and approaches that use evolutionary information, other researchers have proposed foundation models for DNA, such as DNABERT [?]. These models are usually trained on sequences from a single species (most often human) and therefore do not leverage cross-species signals. Some recent methods, such as GPN-MSA [?], do include multiple sequence alignments (MSAs). However, GPN-MSA processes the alignment in a way that loses important information. It treats each column of the alignment as a single input token for its Transformer model. Therefore, the model "flattens" the evolutionary data, ignoring the phylogenetic tree structure that represents how the species within that alignment column are related to each other.

In contrast to the approaches discussed above, our method, GraphyloVar, explicitly learns cross-species dependencies with a graph-based architecture. By incorporating multiple sequence alignments together with their phylogenetic structure from the start of pre-training, it can learn evolutionary constraints across species while preserving the hierarchical relationships between organisms. This design not only improves predictive accuracy, but also maintains a biological relevance by keeping the evolutionary context that many other methods overlook. Building on our previous work with Evolstm [?], where we showed that sequence context strongly influences mutation probabilities, and Graphylo [?], where we showed that incorporating evolutionary information improves the prediction of regulatory sites, we extend these ideas in our new model to capture both the local sequence context and broader evolutionary patterns in an evolution-aware manner.

A key distinguishing feature of GraphyloVar versus GPN-MSA is the explicit modeling of phylogenetic topology. Whereas GPN-MSA flattens the alignment columns into single tokens, GraphyloVar passes per-species features through a Graph Convolutional Network (GCN) whose adjacency matrix encodes the mammalian phylogenetic tree, preserving hierarchical evolutionary relationships.

An important part of the GraphyloVar framework is its pre-training strategy. The model is first pre-trained to predict allele frequencies in human populations, which is a biologically informed pre-training task. This process creates a "foundation" GraphyloVar model that has a general understanding of the evolutionary constraint. This foundation model can then be applied directly to various genomic tasks without task-specific training, or it can be fine-tuned for particular applications. This approach makes the GraphyloVar framework both versatile and comprehensive for predicting the functional consequences of genetic mutations.

Methods

GraphyloVar is a deep learning model developed to predict the functional impact of non-coding variants by modeling their full

sequence and evolutionary context. The model takes as input a 65 bp human DNA sequence centered on the variant of interest, together with orthologous sequences from 57 other mammals and 57 computationally reconstructed ancestral sequences [??]. All sequences are organized by a known phylogenetic tree, so that each observed or reconstructed sequence is associated with a tree node, and their relationships follow the tree topology.

The GraphyloVar architecture is hybrid. Transformer blocks extract context-dependent features from a DNA sequence, capturing context dependencies within the 65 bp window [?]. Graph Convolutional Networks (GCNs) propagate information along the phylogenetic tree, allowing the model to learn how the nucleotide at a position in one species informs the probabilities of nucleotides at the same position in related species. [?]. By combining sequence features with the explicit tree structure, GraphyloVar uses both local sequence context and cross-species information, aiming for more accurate predictions than models that use single-sequence inputs or simple conservation scores.

Data Preprocessing

We utilized multiple sequence alignments (MSAs) from the 100-way whole-genome alignment of vertebrates provided by the UCSC Genome Browser [??]. For our analysis, we extracted sequences for only the 58 extant placental mammalian species. The main reason for this decision is the accuracy of the ancestral sequence reconstruction method. This method is highly accurate for reconstructing ancestral sequences within mammals [?]. Therefore, to ensure the high quality of our input data, we excluded the non-mammalian vertebrate sequences from our analysis.

To reconstruct the evolutionary history of these sequences, we inferred ancestral genome sequences at the internal nodes of the mammalian phylogenetic tree using the Ancestors1.0 program [??]. This maximum-likelihood approach reconstructs the most probable nucleotide at each position for ancestral species based on the sequences of their descendants and the known phylogenetic relationships. The process resulted in a comprehensive dataset comprising 115 sequences per alignment block: 58 from extant species and 57 from inferred ancestors. For consistent model input processing, we segmented genome-wide alignment into standardized blocks of 65 base pairs (bp), with each block centered on the specific genomic position being analyzed. This window size captures sufficient local sequence context while remaining computationally tractable.

Model Architecture

The GraphyloVar architecture consists of two main parts that work in sequence, as shown in Figure 1. First, the Transformer blocks learn features from aligned DNA sequences [?]. Then, the output of the Transformers is passed to a Graph Convolutional Network (GCN), which processes these features using the structure of the phylogenetic tree [?]. These two modules work together to integrate sequence information with broader evolutionary signals.

Embedding and Transformer Blocks

First, we process the aligned DNA sequence for each species in the alignment. Sequences are first one-hot encoded based on a six-letter alphabet (A, C, G, T, -, N). After that, we duplicate each 65-nucleotide aligned sequence into its forward strand and reverse complement, and these two strands are then processed separately in parallel. The one-hot encoded nucleotides in the sequences first go through a dense layer with

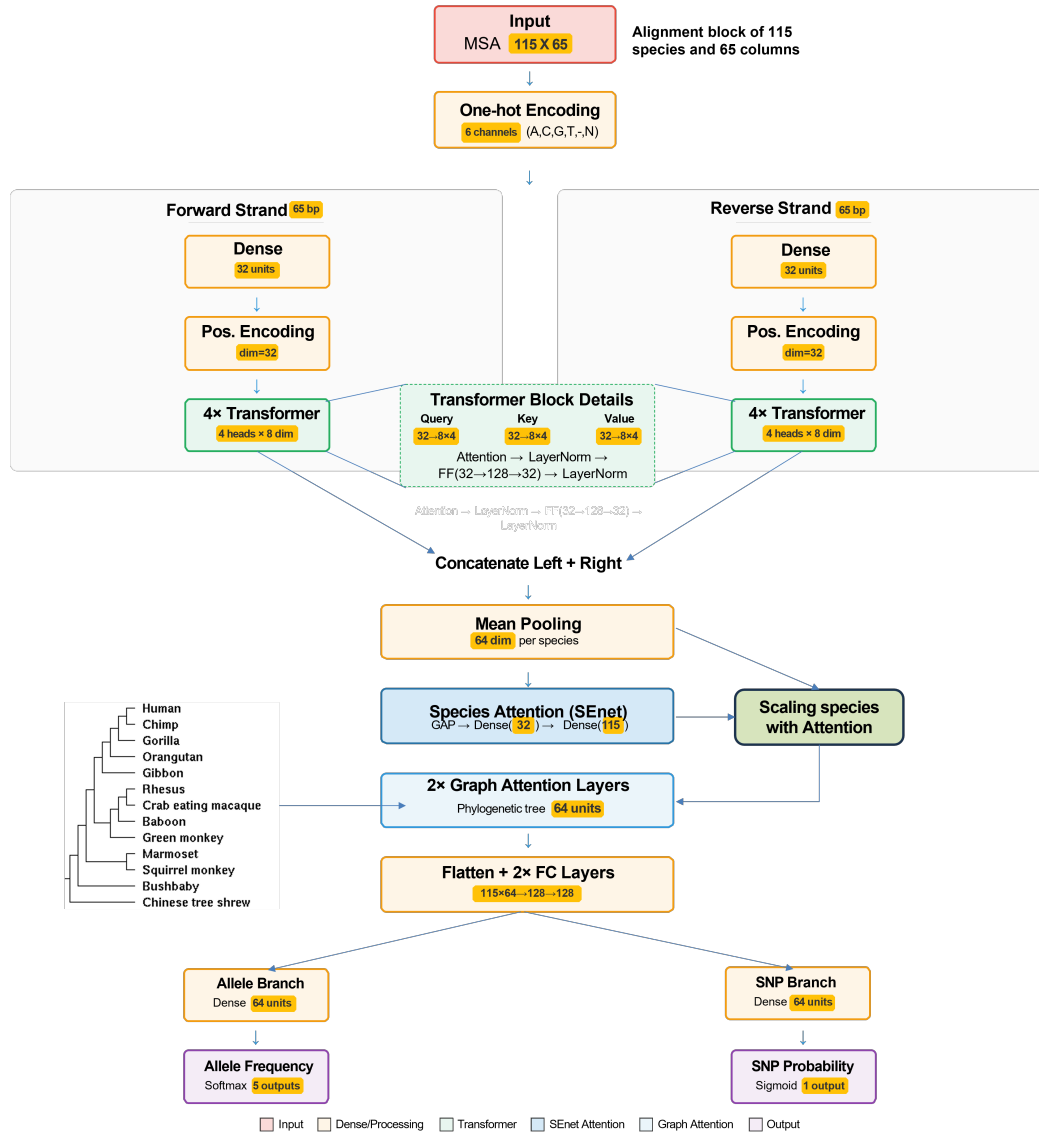


Fig. 1. Overview of GraphyloVar's architecture. For a given 65 bp human sequence and its orthologous and inferred ancestral sequences across 115 placental mammals (58 extant species + 57 ancestral nodes), the sequence is first one-hot encoded and split into forward and reverse strands. It then employs transformer encoder layers for sequence feature extraction. Each strand passes through initial dense layers (32 units) with normalization and GELU activation, followed by positional encoding. Four transformer encoder layers with self-attention mechanisms then process each strand independently, capturing complex sequence patterns and long-range dependencies. The processed representations from both strands are concatenated to form a 64-dimensional feature vector for each position in each species. These features feed into a two-layer Graph Convolutional Network (GCN) that leverages the phylogenetic tree structure to propagate information between evolutionarily related species. The GCN output is flattened and processed through fully connected layers before branching into dual prediction heads: one for allele frequency prediction (softmax activation, predicting probabilities for A,C,G,T,gap) and another for SNP probability prediction (sigmoid activation). This architecture enables GraphyloVar to simultaneously model nucleotide preferences and variation tolerance at each genomic position, providing complementary information for assessing mutation impacts.

32 units. This is followed by batch normalization [?], a GELU activation function [?], and dropout [?]. The main purpose of this embedding is to project the basic one-hot input into a higher 32 dimensional embedding space. This is required to provide the model with a richer, computationally meaningful feature set that allows the internal representation to ensure adequate capacity for the following Transformer attention blocks. Finally, we add a sinusoidal positional encoding to these embeddings. This is an important step because it gives the model information about the position of each nucleotide in the

sequence, which helps the following transformer layers capture position-dependent patterns.

For both the forward and reverse strands, the model applies four Transformer blocks in series [?]. Transformers are suitable for genomic sequence analysis [??]. Their main advantage is that they can use the full sequence context when creating a new, context-aware representation for each position. The key mechanism that allows for this is self-attention, which allows the model to attend to different types of patterns at the same time. After the sequences are processed by the transformers, the final outputs from the forward and reverse

strand pathways are combined. They are first concatenated along the feature dimension. **Rather than using global mean pooling, we extract the representation at the center position of each strand, the position corresponding to the variant of interest. The center features from the forward and reverse strands are concatenated into a 64-dimensional representation for each species. Extracting the center (rather than pooling over the full window) ensures that the model’s downstream prediction is anchored at the variant site while still benefiting from flanking sequence context propagated via self-attention.**

Species Attention and Graph Convolution

We employ a Species attention mechanism, which is a form of Squeeze-and-Excitation (SE) block [?]. The species attention mechanism works by first applying a global average pooling operation to the feature representation of each species. This creates a vector that is then passed through a small multi-layer perceptron (MLP) consisting of two fully connected layers with a sigmoid activation function to produce an attention vector. These attention weights are multiplied by the original feature representations, effectively scaling them based on their importance before they are passed to the GCN layer.

To model evolutionary relationships, GraphyloVar uses a two-layer GCN on the phylogenetic tree [?]. The nodes correspond to extant or ancestral species. Edges indicate evolutionary relationships. The GCN layers work by propagating information between the connected nodes in the tree. This means that the representation for a given species is updated using the information from its direct relatives (its parent and children nodes). This mechanism allows the model to learn about functional elements that are conserved among related species, while still being able to identify patterns that are specific to a single species. The layer update is:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}\right)$$

Here, $H^{(l)}$ is the node-feature matrix at layer l . The matrix $\tilde{A} = A + I$ is the adjacency with self-loops, and $\tilde{D}$ is its degree matrix. The matrix $W^{(l)}$ is learnable. The function σ is the GELU nonlinearity [?]. The second layer outputs a 64-dimensional embedding for each species.

Dual-Output Prediction Heads

After GCN, embeddings of all species are flattened. They pass through two shared fully connected layers with 128 units each. Each layer uses batch normalization, GELU, and dropout. The shared representation then branches into two prediction heads:

- **Allele frequency prediction.** A dense layer with 64 units, followed by batch normalization, GELU, and dropout. A final dense layer with softmax outputs a distribution over five states: A, C, G, T, and a gap/unknown state.
- **SNP probability prediction.** A parallel dense layer with batch normalization, GELU, and dropout. A final sigmoid neuron predicts the probability that a position is a SNP.

This dual-output setup provides complementary signals. The allele head models nucleotide preferences. The SNP head models tolerance to variation. Together, they support downstream assessment of variant effects in a consistent framework.

Model Training and Optimization

GraphyloVar uses a biologically informed pretraining task. We mask only the central human nucleotide in each multiple sequence alignment block. This focuses on learning the position that matters for the variant. Instead of predicting only the reference allele [??], the model predicts the full population-level allele frequency distribution at the masked locus. The training labels come from the Trans-Omics for Precision Medicine (TOPMed) dataset [?], which is a large NIH program with whole-genome sequencing across multiple ancestries. We pre-train the model with the Adam optimizer [?] using the default settings in TensorFlow 2.5 [?]. Mini-batch size is 128. **For the allele frequency head, we use categorical cross-entropy loss, which is appropriate for predicting a probability distribution over the five allele states (A, C, G, T, gap). For the SNP probability head, we use binary cross-entropy loss, since polymorphism status is a binary label.** GELU activation is applied throughout the network for smooth gradients. To limit overfitting, we use dropout with rate 0.2 and early stopping on validation loss.

After the pre-training phase, the model can be used in two primary ways. First, it can be applied directly as a foundation model for zero-shot prediction on various tasks. Alternatively, it can be fine-tuned on specific downstream tasks if labeled data is available. For the task specific fine-tuning step, we updated the weights of the entire model, rather than freezing the pre-trained layers and only training a new classification head.

We tuned the key hyperparameters of the model by evaluating a few candidate values for each parameter while keeping the others constant. The configuration we found was 4 Transformer layers, 2 GCN layers with 32-dimensional node embeddings, and fully connected layers with 128 units. In our experiments, this configuration had a good balance between predictive accuracy and computational cost.

GraphyloVar Score: Quantifying Mutation Impact

GraphyloVar’s dual-output architecture provides two key pieces of information at each genomic position: (1) a probability distribution over the possible alleles $\{A, C, G, T\}$, and (2) the probability that the position is polymorphic (i.e., a SNP). To create a measure of a mutation’s impact, we combine these two outputs into a single score.

For a variant at position i , let $p_a = P(\text{alt}_i \mid \text{MSA}_i, \Theta)$ and $p_r = P(\text{ref}_i \mid \text{MSA}_i, \Theta)$ denote the model’s predicted probabilities for the alternate and reference alleles, and let $p_s = P(\text{SNP}_i)$ denote the predicted probability that position i is polymorphic.

The first component is the **Allele Probability Ratio**, a log-ratio of the alternate- and reference-allele probabilities:

$$\text{Allele Ratio}_i = \log\left(\frac{p_a}{p_r}\right). \quad (1)$$

A lower, more negative score suggests that the alternate allele is much less probable than the reference allele. The model learns this by identifying sequence and evolutionary patterns that suggest the reference allele is under functional constraint. Therefore, a substitution to a low-probability alternate allele is likely to be deleterious.

However, the allele ratio alone does not account for overall tolerance for variation. A site might be highly conserved, meaning any change is likely to be damaging, regardless of the specific alternate allele. To address this, the final **GraphyloVar Score** incorporates the predicted SNP

probability p_s and is defined as:

$$\text{GraphyloVar Score}_i = \log\left(\frac{p_s p_a}{p_s p_r + (1 - p_s)}\right). \quad (2)$$

This combined score offers a more complete picture because if a site is predicted to be highly conserved (p_s is low), the numerator becomes small, resulting in a very low (more deleterious) score for any alternate allele. This correctly reflects that changes at invariant sites are likely harmful. It also retains allele-specific constraint: if a site is predicted to be variable (p_s is high), the score is mainly driven by the ratio p_a/p_r . Lower GraphyloVar scores indicate variants predicted to be more damaging.

Baseline Approaches and Datasets

To evaluate GraphyloVar’s performance, we conducted comparisons against diverse established methods for variant effect prediction. These baseline methods represent a spectrum of approaches, from traditional conservation metrics to advanced machine learning models, each capturing different aspects of genomic functions.

PhyloP [?]

PhyloP measures evolutionary constraint at individual nucleotides by comparing observed substitution rates to neutral evolution. It produces site-specific scores where positive values indicate conservation. Although effective for localized constraints, it lacks a broader sequence context. We utilized 100-way vertebrate PhyloP scores from the UCSC Genome Browser.

PhastCons [?]

In contrast to PhyloP, PhastCons uses a hidden Markov model to identify contiguous conserved elements. It models phylogenetic relationships to detect genomic stretches under purifying selection, making it suitable for regulatory elements that function as cohesive units. We used the 100-way vertebrate PhastCons conservation probabilities.

CADD [?]

CADD integrates genomic annotations into a pathogenicity score using a support vector machine trained to distinguish simulated from observed variants. It combines conservation metrics and functional data to assess potential deleteriousness in both coding and non-coding regions. We used CADD v1.7 scores (GRCh38) for our evaluation.

Enformer [?]

uses a transformer architecture with self-attention to capture long-range interactions (up to 100kb). As the original method does not provide a standard mutation effect score, we trained a simple neural network taking Enformer’s predictions for reference and alternative alleles to predict CADD targets.

GPN-MSA [?]

is conceptually similar to GraphyloVar in using multiple sequence alignments but differs architecturally. It processes concatenated alignment columns through transformers, which does not explicitly preserve phylogenetic relationships.

GPN-Star

GPN-Star [?] is an MSA-aware extension of GPN-MSA, included here as a contemporary MSA-based baseline. The large-scale genomic foundation model Evo2 [?] has inference requirements (compute capability 8.9 or higher, Ada or Hopper

class GPUs) that exceed our current hardware; we therefore discuss it qualitatively as an important future comparison point rather than including it in the quantitative benchmark.

We selected these baselines to represent diverse approaches, including conservation metrics (PhyloP, PhastCons), integrative frameworks (CADD), sequence deep learning (Enformer), and MSA-aware models (GPN-MSA, GPN-Star). While the dbNSFP database contains over 36 algorithms, we focused on these methodologically distinct families to avoid redundancy while still covering the major landscape of variant effect prediction approaches.

To evaluate predictions against experimental data, we compiled a broad collection of Massively Parallel Reporter Assay (MPRA) datasets, covering various regulatory contexts and disease associations. For MPRA studies that focus on enhancer and promoter regions, we incorporated data from studies investigating psychiatric disorders, including schizophrenia [??] and major depressive disorder [?], as well as neurodegenerative conditions such as Alzheimer’s disease and dementia [??]. We also included cancer-related datasets that focus on melanoma [??] and multiple myeloma [?], along with autoimmune conditions such as lupus [?]. To represent expression-modulating variants in non-disease contexts, we included data from the PTGER4 locus study [?]. We also analyzed MPRA studies that focused on variants of oncogenic 3’ UTR mutations associated with prostate cancer [?]. The 3’ UTR MPRA variants were sourced using GWAScatalog annotations to identify disease-associated loci, consistent with the region labels shown in Figures 2 and 3.

Implementation

GraphyloVar was implemented in Python 3.9 using TensorFlow 2.5 on NVIDIA RTX 6000 GPUs. Code is available at <https://github.com/DongjoonLim/GraphyloVar>.

Results

Differentiating Common Variants against rare Variants

To evaluate GraphyloVar’s ability to capture evolutionary constraints that influence allele frequencies, we assessed its zero-shot performance (using the model directly after pre-training without task-specific fine-tuning). Specifically, we tested whether the model could distinguish between common variants (Minor Allele Frequency (MAF) ≥ 0.01) and rare variants (MAF < 0.001) in the human population. We used a chromosome-level holdout strategy: chromosomes 13-22 were excluded entirely from pre-training and validation. All evaluation variants were drawn exclusively from these held-out chromosomes, yielding approximately 149 million held-out SNVs from the TOPMed whole-genome sequencing cohort [?]. This region-based partition eliminates data leakage while ensuring evaluation variants are drawn from a diverse genomic background. This distinction is biologically important, as variants under stronger purifying selection tend to be maintained at lower frequencies in the population.

Figure 2A presents the performance of GraphyloVar compared to established methods (GPN-MSA, GPN-Star, PhyloP, PhastCons, CADD, and Enformer) in different types of genomic contexts. For each method, we calculated the area under the Receiver Operating Characteristics curve (AUROC) to quantify classification performance between common and

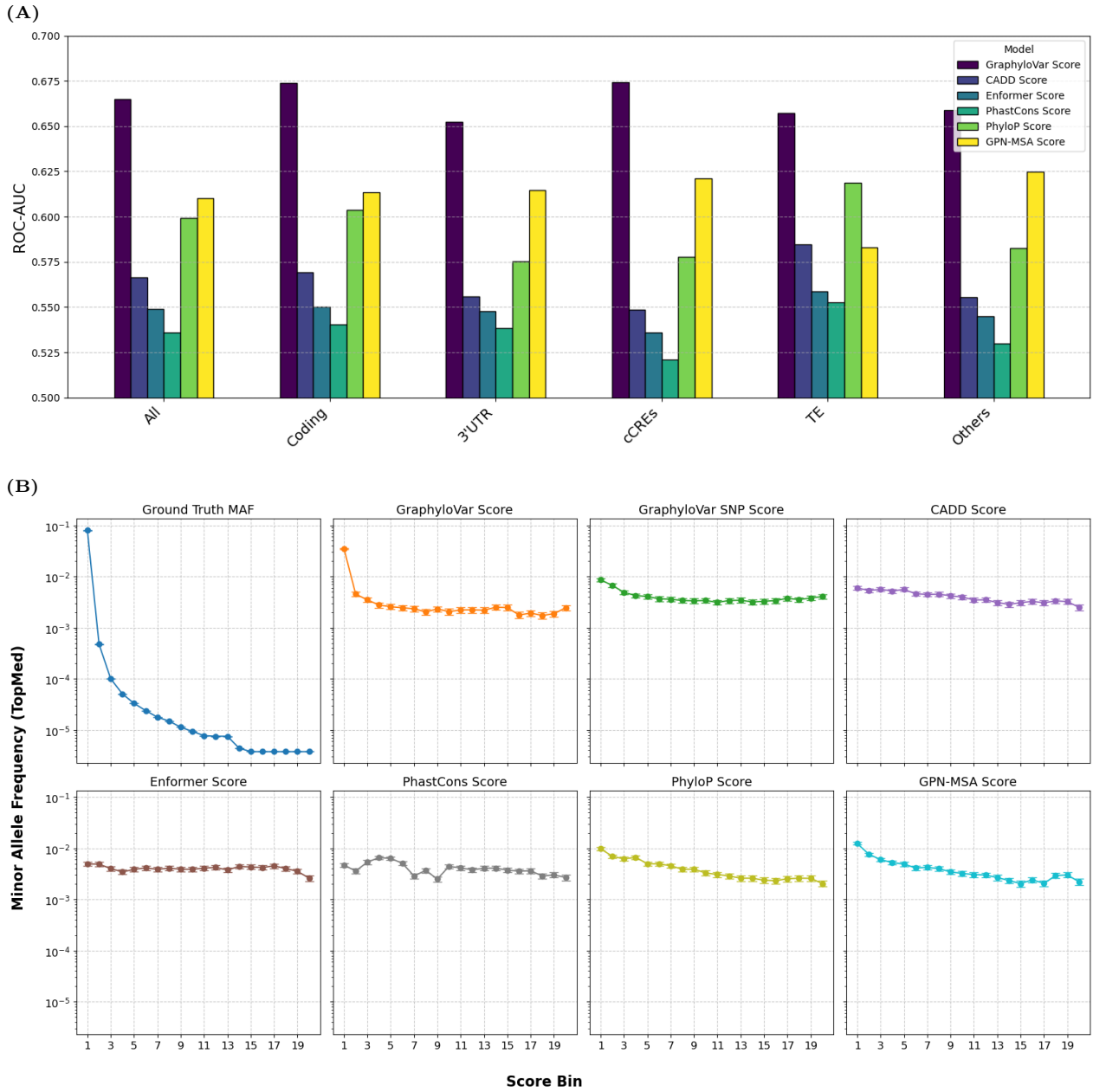


Fig. 2. GraphyloVar's performance distinguishing common from rare variants. **(A)** ROC-AUC scores across genomic region types. The GraphyloVar Allele Ratio Score (dark purple) and GraphyloVar Combined Score (purple) consistently outperform other methods, achieving ROC-AUC ≈ 0.67 in coding regions and cCREs. Traditional conservation methods (PhastCons, PhyloP) and other approaches (CADD, Enformer, GPN-MSA, GPN-Star) score 0.52-0.63. The GraphyloVar advantage is most pronounced in functionally constrained regions. **(B)** Mean MAF (log scale) across variant score bins (1-20). GraphyloVar scores show clearer stratification of variants by frequency than competing methods, with higher-constraint bins corresponding to lower MAF, consistent with purifying selection.

rare variants. The performance advantage was consistent across different genomic contexts, with region-resolved AUROC ≈ 0.67 in coding regions and cCREs and ≈ 0.65 in transposable elements. The overall held-out AUROC on the full chromosome 13-22 test set is 0.6246.

The relationship between predicted scores and actual allele frequencies is further illustrated in Figure 2B, which plots the mean minor allele frequency (log scale) between the score bins for each method. The ground truth panel shows the expected strong inverse relationship between constraint and allele frequency. Among the different prediction

methods, GraphyloVar scores demonstrate clearer stratification of variants by frequency compared to other approaches. Both CADD and Enformer show relatively flat results, indicating a limited ability to discriminate between variants under different levels of selective pressure. PhyloP and GPN-MSA show moderate correlations with allele frequency. This comprehensive performance comparison implies that GraphyloVar consistently outperforms existing methods in different types of genomic regions. This advantage stems from our model's ability to leverage both sequence-specific features

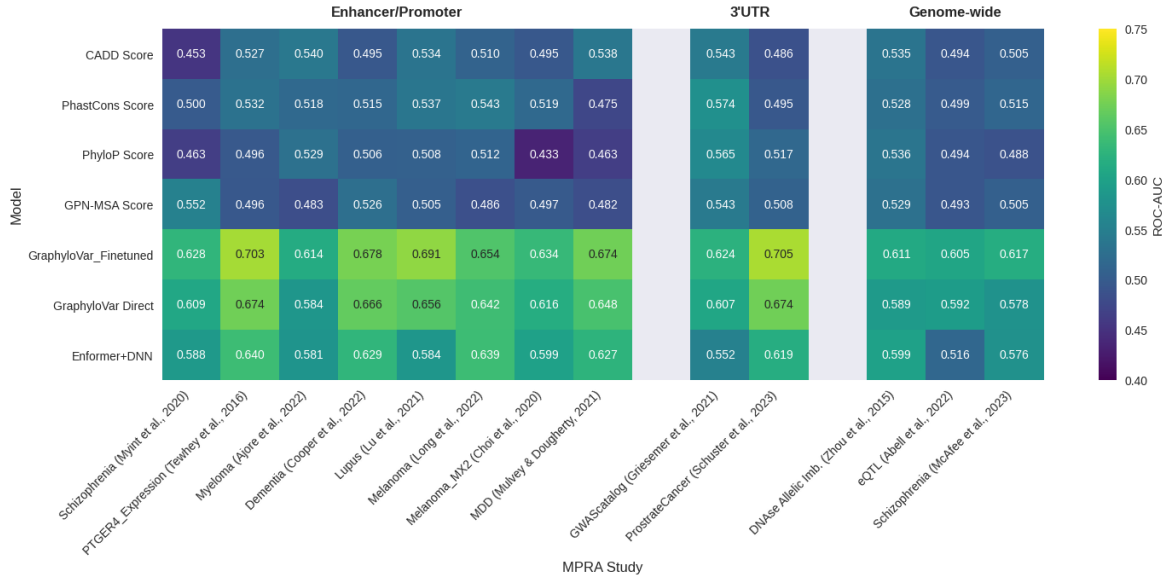


Fig. 3. Heatmap showing ROC-AUC values for different prediction methods (rows) evaluated on 13 MPRA datasets (columns) grouped by types of genomic contexts: enhancer/promoter regions (left), 3' UTRs (middle), and genome-wide analyses (right). The bottom 3 rows show the performance after fine-tuning on MPRA data, directly training GraphyloVar to predict MPRA positive SNPs, and Enformer predictions fed through a DNN to predict MPRA positive SNPs directly.

through Transformer Encoders and phylogenetic relationships through GCNs.

Correlation with Observed Minor Allele Frequencies

To further evaluate how well different methods capture the evolutionary constraints that shape allele frequency distributions, we calculated the Spearman correlation between predicted scores and observed minor allele frequencies on the held-out chromosome 13-22 variant set. GraphyloVar achieved the highest correlation (0.164), outperforming GPN-MSA (0.143), Enformer (0.131), PhyloP (0.099), CADD (0.081), and PhastCons (0.058). This stronger correlation indicates that GraphyloVar's predictions better reflect the underlying selective pressures that determine the frequency of genetic variants in human populations. The improved correlation with MAF demonstrates that GraphyloVar's integration of phylogenetic structure provides more biologically meaningful predictions than approaches that analyze positions independently or rely solely on conservation metrics. By capturing both sequence-specific features and evolutionary relationships, GraphyloVar can better distinguish between variants likely to be tolerated and thus potentially common in the population and those likely to be deleterious and thus maintained at lower frequencies.

Predicting Variant Effects in Massively Parallel Reporter Assays

To evaluate GraphyloVar's ability to predict functional effects of non-coding variants on gene regulation, we benchmarked our model against multiple MPRA datasets spanning diverse regulatory contexts. Figure 3 shows the performance (ROC-AUC) of various prediction methods across 13 different MPRA studies organized into three types of genomic regulatory contexts: enhancer/promoter regions (8 datasets), 3' UTRs (2 datasets), and genome-wide assessments (3 datasets).

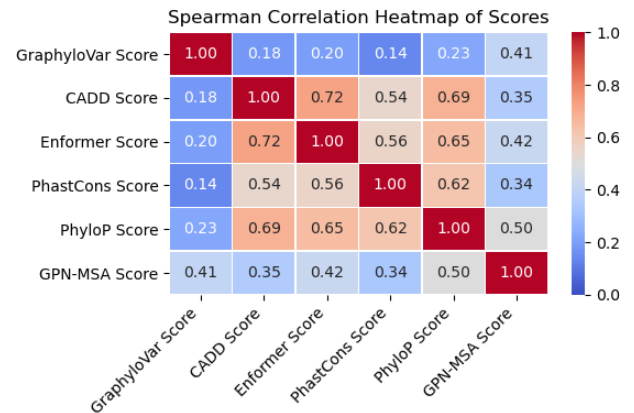


Fig. 4. Spearman correlation among variant scoring methods, calculated on a held-out TopMed SNP dataset that was excluded from the pre-training data.

Performance Improvements Through Fine-tuning

Given GraphyloVar's ability to extract evolutionary signals, we explored two strategies to enhance performance: (1) fine-tuning the pre-trained model on MPRA data, and (2) training a new GraphyloVar model directly on MPRA data without the evolutionary pre-training stage. Additionally, we compared these approaches with a fully connected Deep Neural Network trained on Enformer Scores (see Methods).

GraphyloVar was fine-tuned separately for each type of genomic region and evaluated using a leave-one-out cross-validation approach to mitigate potential training biases. For example, for the enhancer/promoter group (8 MPRA datasets), we conducted eight independent experiments, where in each iteration, one MPRA dataset was designated as the test-set while the model was fine-tuned on the remaining seven.

This strategy ensures that the evaluation reflects the model’s ability to generalize across different datasets within the same regulatory region category. The same strategy was applied to the other two models.

As shown in Figure 3, the fine-tuned GraphyloVar models (Graphylo_finetuned) demonstrated the strongest performance improvements across all MPRA datasets, achieving highest AUC-ROC for 8 out of 13 datasets. In enhancer/promoter regions, this model achieved ROC-AUC values ranging from 0.615 to 0.708. The GraphyloVar models trained directly on MPRA data without pre-training (Graphylo_Direct) and the DNN trained on Enformer scores also showed strong performance, but never as good as Graphylo_finetuned. These findings underscore that while GraphyloVar’s transformer and graph-based architecture inherently provides a robust prediction framework, incorporating experimental data through fine-tuning is paramount for achieving the most precise variant effect prediction. The substantial performance improvements observed through fine-tuning demonstrate that GraphyloVar’s architecture captures core evolutionary signals via its phylogenetic pre-training, which the fine-tuning step then adapts to MPRA-specific regulatory signals.

Complementarity of GraphyloVar Scores with Existing Methods

To assess whether GraphyloVar captures novel and complementary information compared to existing methods, we calculated the Spearman correlation between scores from different predictive approaches. Figure 4 presents a correlation matrix among genomic scoring methods, including GraphyloVar, PhastCons, PhyloP, CADD, and others. Notably, GraphyloVar shows weak-to-moderate correlations with existing methods, indicating that it captures distinct information while still identifying related functional trends. As expected, traditional conservation metrics like PhastCons and PhyloP show stronger correlations with each other than with GraphyloVar, suggesting that GraphyloVar’s phylogenetic modeling extracts signals beyond simple nucleotide-level conservation. In the current Spearman analysis, GraphyloVar remains only modestly correlated with major baselines such as CADD and PhyloP, which supports the interpretation that its signal is at least partially non-redundant.

To further quantify complementarity, we evaluated a z-normalized score ensemble combining GraphyloVar with available partner-model scores. GraphyloVar achieves an overall held-out AUROC of 0.6246 on the full ~149M TOPMed chromosome 13-22 test set. CADD, sign-aligned to the common-versus-rare label convention on this task, achieves ≈ 0.5546 , and PhastCons achieves 0.4667 on the ~5.3M variants for which it has scores. A two-model z-score ensemble of GraphyloVar and CADD achieves an AUROC of 0.6442 on the full held-out variant set, a gain of +0.020 over GraphyloVar alone. Taken together, the low pairwise Spearman correlations and the ensemble AUROC gain support the interpretation that GraphyloVar captures information complementary to existing methods.

To assess the statistical significance of the ensemble gains, we conducted paired two-sided DeLong tests on a uniform random sub-sample of 500,000 held-out variants. The GraphyloVar + CADD ensemble significantly outperforms every individual baseline (vs. GraphyloVar alone: $\Delta\text{AUROC} = +0.0207$, $z = 12.24$, $p < 10^{-15}$; vs. CADD: $\Delta = +0.0950$, $z = 30.62$, $p < 10^{-15}$; vs. PhastCons on the PhastCons-covered subset: $\Delta = +0.1480$, $z = 7.69$, $p = 1.5 \times 10^{-14}$).

Bootstrap percentile 95% confidence intervals ($B = 1000$, sub-sample $n = 200,000$) place the GraphyloVar + CADD AUROC at 0.6496 [0.6395, 0.6587] and GraphyloVar alone at 0.6271 [0.6174, 0.6368], with non-overlapping intervals supporting the ensemble gain.

Context Window Ablation

To justify the default 65 bp context window (32 bp flank on each side of the variant, plus the variant position), we conducted an ablation study evaluating six context-window settings: flank $\in \{0, 1, 8, 16, 32, 100\}$ base pairs, corresponding to input sequences of 1, 3, 17, 33, 65, and 201 bp respectively. All six V3-architecture models were trained on TOPMed chromosomes 1-10 with chromosomes 11-12 as validation, using identical hyperparameters (mixed precision, patience = 30, Adam with ReduceLROnPlateau).

Validation-set nucleotide loss comparisons across flanks show that wider context improves performance up to flank = 32 (65 bp), with diminishing returns at flank = 100 (201 bp) relative to the additional computational cost. We therefore retain the 65 bp window as the default operating point.

GCN Interpretability and Species Importance

Shape Transition from Mean Pooling to GCN.

In the original model, we summarized the species dimension using global average pooling, which collapsed the 115-species alignment into a single feature vector per position. This approach treats every species equally and discards the phylogenetic relationships among them.

In GraphyloVar, we replace mean pooling with a two-layer Graph Convolutional Network (GCN) that operates directly on the phylogenetic adjacency matrix. The input tensor has shape $(B, 115, F)$ where B is the batch size, 115 is the number of nodes (58 extant species + 57 inferred ancestors) in the placental mammalian evolutionary tree, and F is the feature dimension from the upstream Transformer encoder.

The first GCN layer transforms this to $(B, 115, 32)$ by propagating features along edges of the phylogenetic graph, with each node aggregating information from its evolutionary neighbors weighted by the normalized adjacency matrix $\hat{A} = D^{-1/2} A D^{-1/2}$. The second GCN layer further refines these representations to $(B, 115, 32)$. The output is then flattened to $(B, 3680)$ and passed through a shared fully connected layer (128 units, ReLU) before the dual-output prediction heads (allele frequency distribution and SNP probability), each with their own 64-unit dense layer.

This architectural change allows the model to learn species-specific weights that reflect evolutionary importance. An additive perturbation analysis (each species evaluated against an all-gap baseline on 500,000 held-out variants) reveals that 53 of 58 extant placental mammals contribute positive isolated signal, with Human providing the largest contribution ($\Delta\text{CE} = 7.1$). Four diverged primates, Orangutan, Green monkey, Gorilla, and Gibbon, form a secondary cluster ($\Delta\text{CE} = 0.62$ to 1.50), and non-primate mammals contribute a broad gradient ($\Delta\text{CE} = 0.002$ to 0.12). Chimpanzee ranks 11th ($\Delta\text{CE} = 0.062$) despite its closest evolutionary proximity, because its $\approx 98.7\%$ sequence identity with Human yields near-zero marginal (leave-one-out) contribution, reflecting information redundancy with the Human row (see Section 3.6). Five species (Baboon, Rhesus macaque, Marmoset, Squirrel monkey, Jerboa) show marginally negative additive values, likely due to sparse alignment coverage at the tested variant sites. The



Fig. 5. MPRA SNPs prediction accuracy with different alignment coverages.

GCN thus provides an interpretable mechanism for weighting conservation signals across the phylogenetic tree, rather than assuming uniform contribution from all species.

We further investigated species-level importance using a causal perturbation workflow: for each species (or clade), we mask the corresponding alignment rows with a gap token and measure the resulting increase in nucleotide cross-entropy (Δ CE) on held-out variants. A larger Δ CE indicates that the species contributes more information to the model's predictions. Non-primate mammalian clades are evaluated as subtrees (Glires, Cetartiodactyla, Carnivora + Perissodactyla, Chiroptera, Eulipotyphla, Afrotheria + Xenarthra); individual primates and tree shrew are evaluated as single-species leaves.

We evaluated species importance by masking each of the 58 extant placental mammal rows individually on a uniform random sub-sample of 2,000,000 held-out variants drawn across chromosomes 13-22 (200,000 per chromosome). Consistent with the self-row expectation, the ranking is dominated by Human, followed by the closest extant primates (Gorilla, Orangutan, Green monkey, Gibbon); non-primate contributions are positive but orders of magnitude smaller. Full per-species values are shown in Supplementary Figure ??.

The SE-attention gate analysis is reported as a supplementary diagnostic. It reflects the per-species sigmoid gate learned by the Squeeze-and-Excitation block, computed as the mean post-gate feature magnitude averaged over held-out variants. This statistic is correlated with model weighting of species features but is not a causal measure; biological species-importance claims are supported by the perturbation analysis above.

To complement the row-wise species analysis with a within-window view, we also measured position-occlusion saliency across the 66 bp default input window: each column is masked to the gap token across all alignment rows and the resulting Δ CE is measured on the same held-out sub-sample. Results are shown in Supplementary Figure ??.

Ablation Study and Alignment Coverage

Our ablation study confirmed the importance of phylogenetic data. A model using only human sequence performed worst, while adding primate and subsequently full mammalian tree data improved performance. Interestingly, while the model using the full mammalian phylogenetic tree performed the best overall, its improvement over the primates-only model was not as large. This suggests that much of the useful evolutionary signal for this task may come from closely related species. Architecturally, the full GraphyloVar model (Transformers + GCN + Species Attention) consistently outperformed simpler variants.

We also investigated the impact of alignment coverage, defined as the percentage of positions in a 130 bp alignment block containing actual nucleotides rather than gaps (-) or missing data (N) characters. Figure 5 shows the performance of the Graphylo_finetuned and Graphylo_Direct models in three coverage percentile ranges: 0-25% (low coverage), 25-75% (moderate coverage), and 75-100% (high coverage). The results reveal a clear relationship between alignment coverage and predictive accuracy in all genomic contexts. In the lowest coverage percentile (0-25%), both models show substantially reduced performance. This degradation occurs because sparse evolutionary information limits the Graph Convolutional Network's ability to leverage phylogenetic relationships effectively. In particular, performance stabilizes between moderate (25-75%) and high (75-100%) coverage ranges, with similar ROC-AUC values across both percentiles. These findings provide some insights for the GraphyloVar application: regions with very low alignment coverage should be interpreted with caution, while moderate to high coverage regions can be expected to yield reliable predictions.

Computational Time and Scalability

While GraphyloVar’s incorporation of phylogenetic information enhances prediction accuracy, it introduces additional computational complexity compared to single-sequence models. GraphyloVar training required approximately 280 hours on a single NVIDIA Quadro RTX 6000 GPU to complete pretraining on the TOPMed dataset. For prediction, GraphyloVar predicts approximately 20 variants per second on the same hardware.

Discussion and Conclusion

GraphyloVar is a method that integrates deep learning and comparative genomics to predict the effects of mutations in the genome. By using Transformer blocks for sequence feature extraction with graph convolutional networks for phylogenetic modeling, our approach uniquely leverages multi-species alignments to identify evolutionary signals often missed by methods focusing solely on single sequences or conservation metrics.

Our results show that GraphyloVar shows superior performance compared to existing methods. Its pre-trained model effectively distinguishes common from rare variants zero-shot, and after fine-tuning, it accurately predicts regulatory effects measured in MPRA experiments. The model’s superior performance is especially apparent in non-coding regions, where traditional conservation-based approaches often struggle to identify functional elements subject to complex evolutionary dynamics like binding site turnover.

Comparison with PHACT and PHACTboost.

PHACT [?] and its gradient-boosted extension PHACTboost use protein-level features including amino acid conservation, protein structure, and functional domain annotations. These tools are designed specifically for missense variants in protein-coding regions and require protein sequence and structural context as input.

GraphyloVar operates in a fundamentally different scope: it predicts the functional impact of both coding and non-coding variants using only the multi-species whole-genome alignment, without any protein-level features. More than 98% of the human genome is non-coding, and the majority of disease-associated variants identified by GWAS fall in non-coding regions where PHACT and PHACTboost cannot be applied.

The two approaches are therefore complementary rather than competing. PHACT provides high accuracy for missense variant interpretation by leveraging protein context, while GraphyloVar provides genome-wide coverage by learning conservation patterns from the evolutionary tree. For coding variants, a natural extension would be to combine both predictions in an ensemble framework, using PHACT for protein-level features and GraphyloVar for alignment-level phylogenetic features. We leave this interesting combination for future work.

The training strategy based on allele frequency prediction shows some improvement over conventional masked language modeling approaches. By learning to predict population-level variation, GraphyloVar is intended to learn a deep understanding of the evolutionary selective constraints across the genome. This biologically motivated and relevant learning objective achieves effective zero-shot performance without requiring task-specific fine-tuning. GraphyloVar’s dual-output architecture, which predicts both allele frequencies and SNP probabilities, captures complementary aspects of evolutionary

constraint. The primary score effectively identifies substitutions that replace evolutionarily favored nucleotides with disfavored ones, whereas the polymorphism score integrates information about the overall tolerance of variation at each position. This dual perspective corresponds with our comprehension of how natural selection shapes genomic variation through both nucleotide-specific and position-specific constraints.

Despite these advantages, some limitations remain. The reliance on multiple sequence alignments requires extensive computational time compared to single-sequence methods. Additionally, the current implementation focuses on substitutions rather than larger structural variants. In future work, our goal could be to mitigate these constraints through more efficient alignment processing, direct modeling of various types of variants, and integration with functional genomic data to capture tissue-specific effects.

In conclusion, GraphyloVar introduces an approach for variant effect prediction that integrates deep learning with explicit modeling of phylogenetic relationships. Its pre-training strategy on predicting population allele frequencies allows a robust foundation model that captures fundamental evolutionary constraints. Although this foundation provides useful zero-shot capabilities for certain tasks, the model can be effectively fine-tuned on specific experimental data (like MPRA) to achieve high accuracy in predicting functional effects. By leveraging evolutionary information through this pre-training and fine-tuning paradigm, GraphyloVar offers a useful tool to help prioritize functionally relevant variants in a way that complements existing methods.

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Competing interests

No competing interest is declared.

Author contributions statement

D.L. and M.B. conceived the experiments, D.L. conducted the experiments, D.L. and M.B. analysed the results. D.L. and M.B. wrote and reviewed the manuscript.

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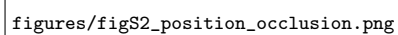
Supplementary Material

Supplementary Figure S1: Per-Species Perturbation
Importance Across All 58 Placental Mammals

Supplementary Figure S2: Position-Occlusion
Saliency

figures/figS6_all58_species_perturbation.png

Fig. S1. Per-species perturbation importance for all 58 extant placental mammals in the GraphyloVar alignment. Each species' alignment row was masked individually to the gap token, and the increase in held-out nucleotide cross-entropy (Δ CE, clipped at zero) was measured on a uniform random sample of 2,000,000 variants drawn across chromosomes 13-22 (200,000 per chromosome). Bars are colored by clade. As expected, Human dominates (the variant position itself sits in the Human row), followed by the closest extant primates (Gorilla, Orangutan, Green monkey, Gibbon); non-primate contributions are positive but orders of magnitude smaller on this sampled subset. This complements the clade-level subtree analysis by providing single-species resolution for every leaf, including non-primate species that are grouped into subtrees in the main analysis.

The figure is a heatmap showing position-occlusion saliency. The x-axis represents the offset from the variant position, ranging from -66 to 66 bp. The y-axis represents the increase in held-out nucleotide cross-entropy (Δ CE). The plot shows a central peak of high saliency at offset 0, with a symmetrical distribution of saliency values on either side. The color scale indicates the magnitude of the saliency, with darker colors representing higher values.

figures/figS2_position_occlusion.png

Fig. S2. Position-occlusion saliency across the default 66 bp input window. Each column of the alignment was masked to the gap token across all species rows, and the resulting increase in held-out nucleotide cross-entropy (Δ CE) was measured on a uniform random sub-sample of 2,000,000 variants drawn across chromosomes 13-22 (200,000 per chromosome). Offset 0 corresponds to the variant position. The pattern indicates which positions within the context window carry the most predictive signal, complementing the external flank-ablation results in Section 3.4.